Department of Electrical Engineering
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Off-school Advanced Learning Programme (2025/26 school year):

Generative AI and Cybersecurity (GenAIC) Engineering Skills for Gifted Students

Fighting F(AI)re with F(AI)re

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Abstract

A deep dive into adversarial machine learning, specifically the AdvGAN and AdvDiff attacks.

Background

- Deep learning models can be easily attacked and fooled;
- Adversarial examples generated using simple attacks like FGSM are easy for humans to spot;
  - ➢ Can we train image generation networks (e.g. diffusion models) to generate high quality, realistic images that fool deep learning models?

Methodology

1. Research different methods of generating images with machine learning models by doing literature reviews on ArXiv;
2. Learn the math behind the machine learning models;
3. Create the models, then implement the adversarial methodology of the academic papers in Google Colab and Kaggle using PyTorch;
4. Train the models, then compare the results by attacking a well-known open source classifier model.

Applications

- Enhancing AI security (e.g. airport security)
- Copyright enforcement

Results

- AdvDiff produces more **high-quality** adversarial examples than AdvGAN, but takes **more time** to generate images after it has been trained;
- Two models were created: An AdvGAN generating L_p norm-restricted adversarial examples trained on the MNIST dataset, and a Diffusion model trained on ImageNet;
- An beginner-friendly guide for GAN/ AdvGAN/DDPM/AdvDiff implementation has been made available.

Source Code

Demo Video

Discussion/Conclusion

- The mathematics/training processes behind AdvGAN and AdvDiff were thoroughly understood;
- My proficiency in PyTorch, namely the nn, functional, and torchvision sub-libraries, has increased;
- Further training of the models using more advanced GPUs would greatly increase the reliability of the results;

References

- AdvGAN by Chaowei Xiao et al.; 2018
- AdvDiff by Xuelong Dai et al.; 2025